# E. Iva & Pav

time limit per test 5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1400 | Tags: binary search, bitmasks, data structures, greedy | https://codeforces.com/problemset/problem/1878/E

*Iva and Pav are a famous Serbian competitive programming couple. In Serbia, they call Pav "papuca" and that's why he will make all of Iva's wishes come true.*

Iva gave Pav an array $a$ of $n$ elements.

Let's define $f(l, r) = a_l \,\&\, a_{l+1} \,\&\, \ldots \,\&\, a_r$ (here & denotes the bitwise AND operation).

**Note that $f(l, r)$ is not defined when $l$ gt $r$.**

Iva also gave Pav $q$ queries.

Each query consists of 2 numbers, $k$ and $l$, and she wants Pav to find the largest index $r$ ($l \le r \le n$), such that $f(l, r) \ge k$.

Pav wants to solve this problem fast because he doesn't want to upset Iva. He needs your help.

## Input

The first line contains a single integer $t$ ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains a single integer $n$ ($1 \le n \le 2 \cdot 10^5$) — the length of array $a$.

The second line of each test case contains $n$ integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^9$) — the elements of array $a$.

The third line of each test case contains a single integer $q$ ($1 \le q \le 10^5$) — the number of queries Iva gave Pav.

The next $q$ lines of each test case contains two numbers, $l$ and $k$ ($1 \le l \le n$, $1 \le k \le 10^9$) — the left bound for the subsegment, and the integer $k$ described in statement.

It is guaranteed that the sum of $n$ over all test cases does not exceed $2 \cdot 10^5$. Also, it is guaranteed that the sum of $q$ over all test cases does not exceed $2 \cdot 10^5$.

## Output

For each query output maximal index $r$ ($l \le r \le n$) such that $a_l \,\&\, a_{l+1} \,\&\, \ldots \,\&\, a_r \ge k$.

If such $r$ doesn't exist, output -1.

## Example

**Input**

```
3
5
15 14 17 42 34
3
1 7
2 15
4 5
5
7 5 3 1 7
4
1 7
5 7
2 3
2 2
7
19 20 15 12 21 7 11
4
1 15
4 4
7 12
5 7
```

**Output**

```
2 -1 5
1 5 2 2
2 6 -1 5
```

## Note

In the **first test case** $n$=5, and the array $a$ = [15, 14, 17, 42, 34]

The first query asks for the biggest index $r$ such that the $f(1, r) \geq 7$.

$f(1,1) = 15$, $f(1, 2) = 14$, $f(1,3)=0$ $f(1, 4)=0$ $f(1, 5)=0$, so $r$=2 is the answer.

The second query asks for $f(2, r) \geq 15$. Since such $r$ doesn't exist, the answer is -1.

The third query asks for $f(4, r) \geq 5$. $f(4, 4) = 42$, $f(4, 5) = 34$, so $r$=5 is the answer.

In the **second test case** $n$=5, and the array $a$= [7, 5, 3, 1, 7].

For the first query, $f(1, r) \geq 7$.

$f(1, 1)=7$, $f(1, 2)=5$, $f(1, 3) = 1$, $f(1,4) = 1$, $f(1, 5)=1$, so the answer to this query is 1.

For the second query, $f(5, r) \geq 7$.

$f(5, 5) = 7$, so the answer is 5.

For the third query, $f(2, r) \geq 3$.

$f(2, 2) = 5$, $f(2, 3) = 1$, $f(2, 4) = 1$, $f(2, 5) = 1$, so the answer is 2.